~~1, 2, 3, 4~~ $k = 5$

$1+4 = 5$ $= 2$

~~3~~, 1, ~~3~~, 4, 3 $k = 6$

$= 4$

brute force 1, 2, 3, 4

i=1 1+2 / 1+3 / 1+4 = 5 cont++

i=2 2+3 / 2+4 cont++

5

$O(n^2)$

for i = 0 → n

for j = i+1 → n

it is slow because we are doing

↳ * hash map cost

* sorting

?

1) Sort , 3, 1, 3, 4, 3 $k = 6$

sorted: [1, 3, 3, 3, 4]

L R

smallest largest

sum

cage R

case 1	case 2	case 3
sum == k	sum < k	sum > k
found	L is small	R is too big
move both	move L →	move R ←
1, 3, 3, 3, 4	1+4 = 5 < 6 L →	
L R	7 > 6 R ←	
L R	6 == 6 cont++	
R	L meet R → stop	

Complexity

$\boxed{\text{sort} = O(n \lg n)}$

$O(n)$

Hash map O(1)

for i = 0 → n
  for j = i+1 → n
    if nums[i] = nums[j] = k → count++

search for complement
k − nums[i]
→ needs to be stored in map

O(1)

2 | 4 | 0 = 12dt

vacant room

tick = 4 dt

person (col 1)
1) 4
2) 2

count++

number arrive
complement in room? Yes: pair up, both leave
No sit down and wait

k = 6

number arrive	needed complement	found in room	room {}
3	3	no	3:4
4	5	no	3:4, 4, 4
3	3	yes count	~~3:0 : 4, 4~~
4	2	no	1:1 ; 2:4
3	3	no	✓

1

Complex O(n)